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# Probing diametric coordination graphs for multi-agent reinforcement learning

Mutong Liu[a,b], Tiantian He[c], Yang Liu[a,b,*], Jiming Liu[a,b], Yew-Soon Ong[c,d]

[a] *Department of Computer Science and Centre for Health Informatics, Hong Kong Baptist University, Hong Kong SAR, China*
[b] *Institute for Research and Continuing Education, Hong Kong Baptist University, Shenzhen, China*
[c] *Center for Frontier AI Research, Institute of Advanced Intelligence and Computing, Agency for Science, Technology and Research, Singapore*
[d] *College of Computing and Data Science, Nanyang Technological University, Singapore*

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A B S T R A C T

Effective coordination is the cornerstone of success in cooperative Multi-Agent Reinforcement Learning (MARL), where agents work together to achieve a collective goal. However, most existing methods for modeling agent coordination rely primarily on proximity between agents' features or on incorporating heterogeneity at the behavioral/policy level, while overlooking the valuable diversity present at the observation level. In this paper, we introduce Diametric Coordination Graphs (DiaCoG), a novel framework that dynamically models implicit coordination of agents by integrating both consistency (shared similarities) and discrepancy (unique differences) from agents' observations, offering a richer understanding of inter-agent relationships. Accordingly, we propose two approaches, DiaCoG-DE and DiaCoG-CE, implementing this framework under two MARL architectures of actor-critic networks, Centralized Training and Decentralized Execution (CTDE) and Centralized Training and Centralized Execution (CTCE), respectively. Through theoretical analyses based on information-theoretic measures, we show that DiaCoG offers superior expressiveness over consistency-based methods for value estimation and action selection. Empirically, we evaluate the proposed methods in the Predator-Prey and Traffic Junction environments, where they outperform baselines in terms of both final average returns/success rates and convergence speed across diverse scenarios. To further showcase the adaptability of DiaCoG across different MARL paradigms, we integrate it with a state-of-the-art value-based approach and compare it against several representative and state-of-the-art methods to demonstrate the enhanced coordination performance by DiaCoG in the StarCraft II Multi-Agent Challenge environment.

## 1. Introduction

Multi-agent reinforcement learning (MARL) in cooperative scenarios focuses on developing optimal strategies that enable multiple intelligent agents to collaboratively achieve a common goal [1–3]. This has proven valuable in tackling complex real-world challenges, including coordinating drone swarms for search-and-rescue missions [4], enabling autonomous driving [5], managing multiple robots for diverse tasks [6], and optimizing traffic light coordination [7]. In these complicated cooperative settings, effective coordination among agents is essential for achieving collective success, which means agents need to interact and cooperate with each other to foster

* Corresponding author.
*E-mail address:* csygliu@comp.hkbu.edu.hk (Y. Liu).

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coherent joint behaviors, so as to enhance overall outcomes [8,9]. Thus, designing learning mechanisms that enable informative and efficient coordination to enhance decision-making is a critical priority in MARL.

Efforts to address the coordination challenge in cooperative MARL have primarily followed several research directions. One line of work decomposes the joint value function into local components to determine the optimal joint action effectively. These methods either rely on predefined graph structures to explicitly model agent interactions [10,11] or learn implicit coordination relationships dynamically [12–16]. Agent communication and relation modeling represent another prominent line of work for enhancing coordination. This class includes methods that enable agents to share messages via explicit communication protocols [17–20], as well as approaches that learn collaboration via implicit relational graphs [21–24]. A third line of research explores the role of behavioral diversity [25,26] and agent heterogeneity [27–29]. These approaches foster coordination by deliberately enforcing or inducing differences among agents either at the policy/action level (e.g., via diversity objectives) or at the agent/role level (e.g., via heterogeneous policy network designs), thereby encouraging role specification towards shared goals.

While these methods have significantly advanced coordination in MARL, many still have certain limitations. Early approaches often rely on static, pre-defined topologies [10]. More recent methods that learn dynamic and implicit coordination structures typically follow a proximity/similarity-based intuition, assuming that agents with similar observations tend to cooperate closely. Common examples include leveraging Euclidean distance between agents' positions [20], neighborhood cognitive consistency [24], or dot-product attention mechanisms [16,22] to infer the coordination structures. In parallel, some methods introduce heterogeneity into multi-agent systems at the algorithmic or mechanism level [25–28]. However, most existing work has overlooked the potential of explicitly capturing discrepancy information arising from diverse agents at the observation level, in addition to consistency. This specific focus limits the adaptability and generalizability of current methods in complex and dynamic multi-agent environments.

The overlooked potential of discrepancy or dissimilarity in agent observations presents a complementary opportunity to advance cooperative MARL, specifically on implicit coordination relation modeling. Discrepancy or dissimilarity can reveal unique agent context, such as different positions, roles, or environmental cues, which are essential for complementary actions in diverse scenarios. Drawing inspiration from broader machine learning and information sciences fields, where leveraging such diversity in data and representations has been shown to improve outcomes [30–32], the integration of consistency and discrepancy information can also be instrumental in addressing the coordination challenge in MARL. For example, in collaborative scenarios like the Predator-Prey introduced in Section 5.1, predators with similar observations are often physically proximate, resulting in significant overlap in local observations. In such cases, consistency-based methods naturally prioritize the collaboration between the agent and its neighbors with the most similar observations. However, this overlap may offer limited additional value beyond individual observations. Conversely, information from agents with dissimilar observations–reflecting diverse perspectives or roles–can provide substantial benefits for learning a more informed policy, as evidenced by complementary actions that enhance team coordination, as discussed in Section 6.

To address this gap, we propose Diametric Coordination Graph (DiaCoG), a novel framework that dynamically learns implicit coordination structures by integrating both consistency and discrepancy information from agent observations, as illustrated in Fig. 1. Unlike prior methods that predominantly focus on proximity between agents, DiaCoG leverages the complementary nature of these dual perspectives to capture richer inter-agent relationships, enabling more effective collaboration. This strategy enhances the informativeness of agent interactions by explicitly incorporating the diversity and prioritizing coordination with the most relevant agents. By doing so, DiaCoG empowers MARL with the adaptivity needed for complex cooperative tasks, offering a significant ingredient for achieving collective goals. Based on the concept of DiaCoG, we further develop two approaches, DiaCoG-DE and DiaCoG-CE, under two MARL architectures of actor-critic networks, i.e., Centralized Training and Decentralized Execution (CTDE) and Centralized Training and Centralized Execution (CTCE), respectively.

We validate the effectiveness of DiaCoG-DE and DiaCoG-CE through both theoretical and empirical investigations. Theoretically, we analyze its enhanced expressiveness for policy learning over consistency-based methods under CTDE and CTCE architectures, respectively. Using information-theoretic tools such as mutual information and entropy, we demonstrate how discrepancy information augments value estimation in the CTDE architecture and action selection in the CTCE architecture. Empirically, we evaluate DiaCoG in two typical cooperative MARL environments, i.e., Predator-Prey and Traffic Junction, comparing it against several baselines. Our results show that DiaCoG achieves faster convergence and higher final returns/success rates in most scenarios, validating its effectiveness in cooperative MARL. To further illustrate the adaptivity of the proposed framework across different MARL paradigms, we integrate DiaCoG with a state-of-the-art (SOTA) value-based algorithm, Group-Aware Coordination Graph (GACG) [15]. We then evaluate its effectiveness in the StarCraft II Multi-Agent Challenge (SMAC) environment against several representative and SOTA methods. This integration enhances coordination effectiveness, thus showing DiaCoG's flexibility across varied architectures.

In summary, this work makes the following contributions:

- We introduce Diametric Coordination Graph (DiaCoG), a novel concept that dynamically learns implicit coordination structures by integrating consistency and discrepancy information from agent observations, enabling richer and more adaptive collaboration in MARL.
- Building upon the proposed concept, we develop two approaches, DiaCoG-DE and DiaCoG-CE, tailored for the CTDE and CTCE architectures of actor-critic networks, respectively, demonstrating DiaCoG's flexibility and seamless integration into existing MARL frameworks.
- We provide theoretical analyses of DiaCoG's expressiveness, showing how dual information enhances value estimation under CTDE and action selection under CTCE over consistency-based methods. These analyses reveal DiaCoG's capacity to capture richer information for the complex dependencies between agents to enhance coordination behaviors.

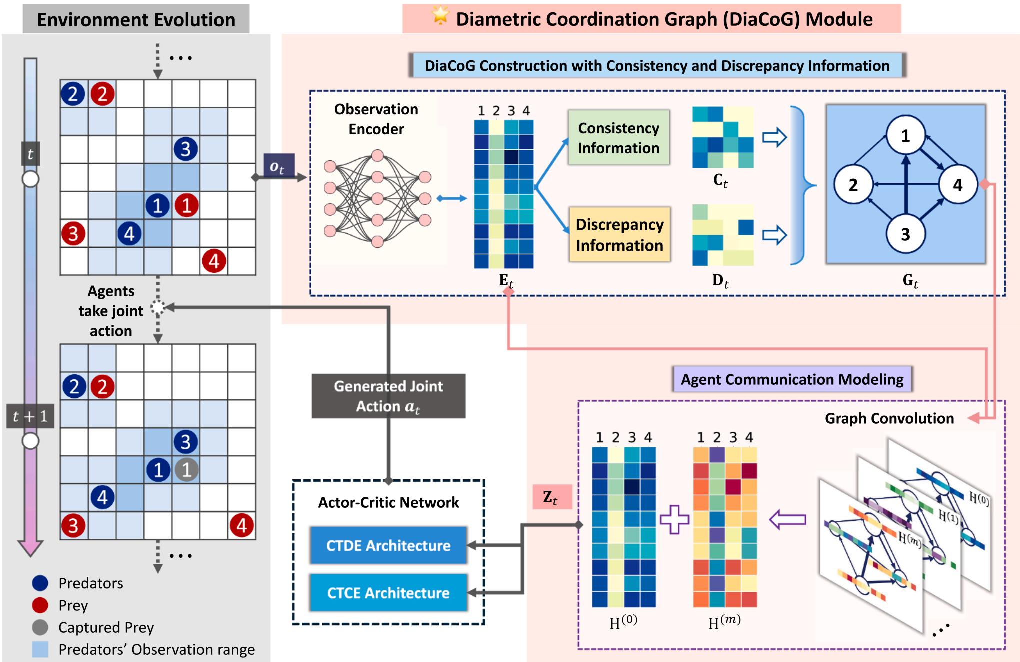



**Fig. 1.** Overview of the DiaCoG framework. The DiaCoG framework integrates two key components to enhance multi-agent coordination. The first component extracts consistency and discrepancy information from feature embeddings derived by encoding agents' raw observations (illustrated with a grid-based environment, such as the Predator-Prey scenario, on the left) and constructs the DiaCoG structure to model inter-agent relationships using this extracted information. The second component models agent communications by applying graph convolution to agents' feature embeddings and the DiaCoG structure, generating enriched graph representations. These representations of agent communications are subsequently used by actor-critic networks for value estimation in the CTDE architecture and action generation in the CTCE architecture.

- We empirically evaluate DiaCoG across a series of scenarios (i.e., varying collaboration penalty settings and difficulties) in two cooperative MARL environments. The advantage of DiaCoG over several baselines, in terms of convergence speed and final outcomes, confirms its effectiveness. To further demonstrate the adaptability of DiaCoG, we integrate it with a SOTA value-based method in MARL and validate its consistently superior performance over several representative and SOTA methods in the SMAC environment.

The remainder of this paper is organized as follows. Section 2 details our proposed concept and methods. Section 3 presents the theoretical analyses. Section 4 describes the overall experimental setup and baselines for comparison. Specific environmental settings and experimental results for the Predator-Prey environment are presented in Section 5, while a case study based on this environment illustrating how DiaCoG enhances coordination behaviors is given in Section 6. Environmental settings and experimental results for the Traffic Junction and SMAC environments are presented and discussed in Sections 7 and 8, respectively. Section 9 reviews related literature. Finally, Section 10 concludes the paper and discusses the potential applications and limitations of our work, as well as future directions.

## 2. Diametric Coordination Graphs for MARL

This section presents Diametric Coordination Graphs (DiaCoG), a novel framework that leverages both consistency and discrepancy information from agent observations to dynamically model implicit coordination structures. We begin with the fundamentals of MARL and the Multi-Agent Proximal Policy Optimization (MAPPO), which provide the necessary preliminaries for the proposed approach. Building on this basis, we then detail the DiaCoG methodology, including the graph structure generation with consistency and discrepancy information, agent communication modeling, and policy learning under CTDE and CTCE frameworks.

### 2.1. Preliminaries of MARL

The cooperative MARL problem explored in this paper is formalized as a Decentralized Partially Observable Markov Decision Process (DEC-POMDP) [33], which is defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{O}, R, \mathcal{P}, n, \gamma \rangle$. Here, $\mathcal{S}$ denotes the state space of the environment, with $s \in \mathcal{S}$ representing the global state; $\mathcal{A}$ is the joint action space, and $\boldsymbol{a} = (a_1, \dots, a_n) \in \mathcal{A}$ denotes the joint action for all agents; $\boldsymbol{o} = \mathcal{O}(s) = \{o_i \mid i = 1, \dots, n\}$ denotes the set of local observations for all agents under the global state $s$, where $o_i$ is the local observation for agent $i$; $n$ denotes the number of agents; $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function, with $r = R(s, \boldsymbol{a})$ being the immediate collective/team reward obtained from the environment by deploying the joint action under the global state $s$; $\mathcal{P}(s'|s,\boldsymbol{a})$ denotes the transition probability from $s$ to $s'$ given the joint action $\boldsymbol{a}$; and $\gamma \in [0,1]$ is the discount factor, with the future cumulative discounted return at time step $t$ given by $G_t = \sum_{l=0}^{\infty} \gamma^l r_{t+l}$. Each agent $i$ follows a stochastic policy $\pi_{i,\theta}(a_i|o_i)$, parameterized by $\theta$, to select action $a_i$ based on its local observation $o_i$. The primary objective of policy learning is to maximize the expectation of discounted accumulated return by optimizing the policies of all agents:

$$\max_{\pi_1,\ldots,\pi_n} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R\left(s_t, \boldsymbol{a}_t\right)\right]. \tag{1}$$

### 2.1.1. MAPPO

To facilitate the investigation of our DiaCoG framework, we adopt the Multi-Agent Proximal Policy Optimization (MAPPO) [34], a widely used policy gradient MARL method, to maximize the discounted accumulated return. In the single-agent situation, the Proximal Policy Optimization (PPO) algorithm [35] employs an actor-critic architecture with two separate neural networks: a policy function (actor) $\pi_\theta$ generating actions; and a value function (critic) $\hat{V}_\xi(s)$ providing the value estimation of the current state.

For policy optimization, MAPPO maximizes PPO's clipped surrogate objective:

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\left[\min(ratio_t(\theta)\hat{A}_t, \mathrm{clip}(ratio_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)\right], \tag{2}$$

where $ratio_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t|s_t)}$ represents the probability ratio between the updated policy and the old policy, which is used to control the policy update magnitude; $\epsilon$ is a hyperparameter to constrain the update interval; and $\hat{A}_t$ denotes the estimate of the advantage value at time step $t$, which is calculated using the truncated Generalized Advantage Estimation (GAE). To encourage exploration, MAPPO incorporates an entropy bonus of the current policy, i.e., $B[\pi_\theta](s_t)$, which maximizes the entropy of the action distribution generated by $\pi_\theta$ under state $s_t$. Therefore, the overall objective function for policy optimization is:

$$L_t^{PF}(\theta) = \hat{\mathbb{E}}_t\left[L_t^{CLIP}(\theta) + cB\left[\pi_\theta\right]\left(s_t\right)\right], \tag{3}$$

where $c$ is a weight coefficient balancing the trade-off between policy improvement and exploration.

To optimize the value function $\hat{V}_\xi(s)$ for accurately predicting the state value at time step $t$, denoted as $v_t^{target}$ and defined as the true cumulative discounted reward from time $t$ onward, we adopt the maximum likelihood estimation with the following objective function:

$$L_t^{VF}(\phi) = \hat{\mathbb{E}}_t\left[\log \hat{V}_\xi(v_t^{target} \mid s_t)\right]. \tag{4}$$

Under the assumption of a Gaussian distribution for the value estimates, the critic network outputs the mean $\mu_\phi(s_t)$ as the predicted state value at time step $t$, i.e., $\hat{v}_t = \mu_\phi(s_t)$.

The actor and critic architectures are tailored to the specific requirements of CTDE and CTCE schemes, with detailed configurations provided in Section 2.2.3.

## 2.2. Diametric Coordination Graphs

Rather than relying solely on proximity-based measures to infer dynamical coordination relationships between agents, we employ a more comprehensive approach by integrating both consistency and discrepancy information from agent observations to improve the effectiveness of coordination modeling. The pseudocode of the proposed DiaCoG method is presented in Algorithm 1.

### 2.2.1. DiaCoG construction with consistency and discrepancy information

We first compute the consistency $\mathbf{C}_{ij}$ and discrepancy $\mathbf{D}_{ij}$ for agent pairs $(i,j) \in 1,\ldots,n$ from their observations' embeddings. For agent $i$, its embedding for subsequent processing is derived by encoding its raw observations with an encoder: $\mathbf{E}_{i,:} = \mathrm{Encoder}(o_i)$. Then, the $\mathbf{C}_{ij}$ and $\mathbf{D}_{ij}$ are extracted using the following equations:

$$\begin{aligned}
\mathbf{C}_{ij} &= \mathbf{A}_{ij} \cdot \left[\mathbf{a}\left(\mathbf{E}_{i,:}||\mathbf{E}_{j,:}\right)^T + \mathbf{P}_{i,:}\,\mathrm{diag}(\mathbf{b})\mathbf{P}_{j,:}^T\right],\\
\mathbf{D}_{ij} &= \mathbf{A}_{ij} \cdot \left[\alpha\left|\mathbf{E}_{i,:} - \mathbf{E}_{j,:}\right|^2 + (1-\alpha)\left|\mathbf{P}_{i,:} - \mathbf{P}_{j,:}\right|^2\right],
\end{aligned} \tag{5}$$

where $\mathbf{A}$ denotes the adjacency matrix of agents, $\mathbf{P}$ is the latent position matrix, vectors $\mathbf{a}$ and $\mathbf{b}$ are learnable parameters, the operator $||$ denotes vector concatenation, $\mathrm{diag}(\cdot)$ converts a vector to a diagonal matrix, $|\cdot|$ denotes the $l_2$-norm of the vector, and $\alpha \in [0,1]$ is a learnable parameter balancing the relative importance of two items.

In real-world scenarios such as autonomous driving and robots' collaboration, the topologies of agent interactions are often dynamic and unknown. Therefore, we assume $\mathbf{A} = \mathbf{1}_{n\times n}$ (an all-one matrix), indicating potential connectivity among all agents, with specific coordination structures inferred later. For scenarios with predefined topologies, the matrix $\mathbf{A}$ can be adjusted to match the problem context. The latent position matrix $\mathbf{P} \in \mathbb{R}^{n\times d}$ typically encodes structural position information of nodes in a graph to represent the neighborhood relationships among agents. Since the fixed and known global graph structure is often unavailable in most practical scenarios, we construct $\mathbf{P}$ via selecting the top-$K$ nearest neighbors for each agent based on a general dot-product attention on agent

**Algorithm 1:** Diametric Coordination Graphs (DiaCoG).

**Input:** Raw observations of all agents $\boldsymbol{o}$ at one time step
**Output:** The joint action $\boldsymbol{a}$ at one time step
Encode observations of $n$ agents: $\mathbf{E}_{i,:} \leftarrow \text{Encoder}(o_i), i = 1, \ldots, n$;
Calculate the position matrix $\mathbf{P}$ based on $\mathbf{E}$;
Map encoded observations and position matrix by projection matrices: $\mathbf{E} \leftarrow \mathbf{E}\mathbf{W}_e$, $\mathbf{P} \leftarrow \mathbf{P}\mathbf{W}_p$ ;
Extract consistency and discrepancy matrices, $\mathbf{C}$ and $\mathbf{D}$, by Eq. (5);
Generate the DiaCoG structure $\mathbf{G}$ based on $\mathbf{C}$ and $\mathbf{D}$ by Eq. (6);
$\mathbf{H}^{(0)} \leftarrow \mathbf{E}$;
**for** $l = 0$ *to* $m-1$ **do**
  Conduct the $l$-layer graph convolution by Eq. (7) to obtain $\mathbf{H}^{(l+1)}$;
**end**
Generate DiaCoG embeddings by residual link: $\mathbf{Z} \leftarrow \mathbf{H}^{(0)} + \mathbf{H}^{(m)}$;
**if** *CTDE* **then**
  Input raw observations $o_i$ into policy network $\pi_{i,\theta}$ to generate action $a_i$ for each agent $i$;
  Input $\mathbf{Z}$ into value network $\hat{V}_\xi$ to estimate the state value;
**end**
**if** *CTCE* **then**
  Input $\mathbf{Z}_{i,:}$ into policy network $\pi_{i,\theta}$ to generate action $a_i$ for each agent $i$;
  Input $\mathbf{Z}$ into value network $\hat{V}_\xi$ to estimate the state value;
**end**
Deploy joint action $\boldsymbol{a}$ to environment to obtain immediate reward $r$ and observations for the next time step;
**return** *Joint action $\boldsymbol{a}$ at one time step*

observation embeddings[1], so as to generically represent neighborhood relationships. Specifically, we set $d = n$ in our implementation, and for each agent $i$, the $i$-th row of $\mathbf{P}$ reflects the closeness of the top-$K$ most similar agents (measured via a general dot-product attention mechanism over $\mathbf{E}$), with values of 0 for agents outside the top-$K$ range.

Following the extraction of consistency and discrepancy information from agent embeddings, we calculate the structure of the diametric coordination graph, $\mathbf{G}$, as follows:

$$\mathbf{G}_{ij} = \mathbf{A}_{ij} \cdot \exp\left(\mathbf{C}_{ij} - \beta \mathbf{D}_{ij}\right), \tag{6}$$

where the learnable parameter $\beta$ balances the influence of consistency and discrepancy information, and $\exp(\cdot)$ applies an exponential function to shape the graph structure with a sparse property.

### 2.2.2. Agent communication modeling

Building on the constructed DiaCoG structure, we model agent communications using Graph Convolutional Networks (GCNs), which are widely adopted due to their message-passing and aggregation capabilities. In each layer of the $m$-layer GCN, the information exchanged between agents is computed by the following equations:

$$\begin{aligned} \mathbf{M} &= \widetilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \widetilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)}, \\ \mathbf{H}^{(l+1)} &= \sigma(\mathbf{M}), \end{aligned} \tag{7}$$

where $\mathbf{M}$ is the matrix of agents' integrated graph messages after the graph convolution, $\tilde{\mathbf{A}} = \mathbf{G}$ (the DiaCoG structure, including self-connections due to non-zero diagonals), and $\widetilde{\mathbf{D}}$ is the degree matrix with diagonal elements $\widetilde{\mathbf{D}}_{ii} = \sum_{j \in \mathcal{N}_i} \tilde{\mathbf{A}}_{ij}$ ($\mathcal{N}_i$ is the neighbor set of agent $i$). The activation function $\sigma(\cdot)$ is implemented as the Rectified Linear Unit (ReLU) to enable non-linear feature transformation. The initial embedding matrix is set to $\mathbf{H}^{(0)} = \mathbf{E} = [e_1^\top, \ldots, e_n^\top]$, which is the encoded embeddings derived from agents' raw observations $(o_1, \ldots, o_n)$. The graph convolution operation will repeat $m$ times and generate a series of integrated graph embeddings $\{\mathbf{H}^{(1)}, \ldots, \mathbf{H}^{(m)}\}$. Follow the setting of residual link in [22], we set the final graph embeddings $\mathbf{Z}$ by integrating the initial embeddings $\mathbf{H}^{(0)}$ with the last graph embeddings $\mathbf{H}^{(m)}$, i.e., $\mathbf{Z} = \mathbf{H}^{(0)} + \mathbf{H}^{(m)}$. These embeddings are subsequently used for value estimation in the CTDE architecture and action selection in the CTCE architecture.

### 2.2.3. Policy learning with DiaCoG embeddings

In this paper, we introduce two variants, DiaCoG-DE and DiaCoG-CE, adapting the DiaCoG to the CTDE and CTCE architectures, respectively, to enhance coordination performance with actor-critic networks, as illustrated in Fig. 2. The DiaCoG embeddings of

[1] Note that using "top-$K$ most similar agents" to construct $\mathbf{P}$ does not limit us to similarity-only coordination. Based on Eq. (5), we naturally obtain a measure of dissimilarity for neighborhood relations: agents whose neighborhood profiles differ substantially (i.e., large $|\mathbf{P}_{i,:} - \mathbf{P}_{j,:}|^2$) receive high discrepancy scores and therefore have the chance to be strongly connected in the coordination graph when they carry complementary information.

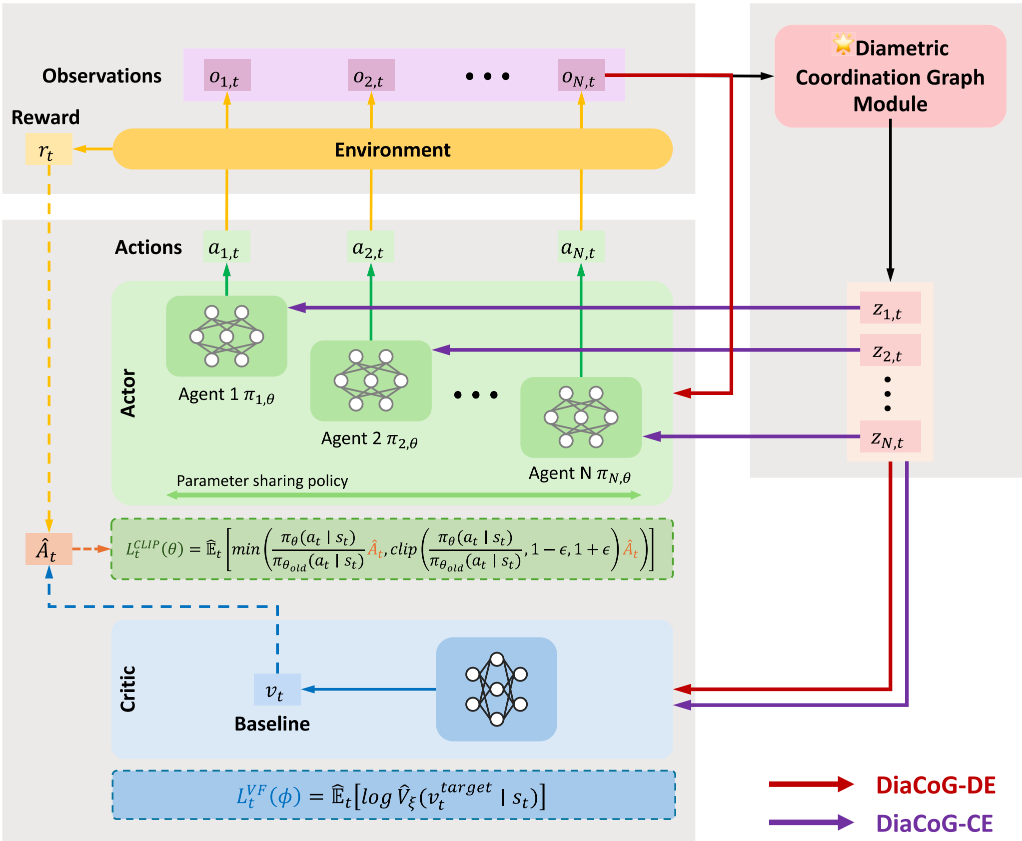



**Fig. 2.** Implementation of DiaCoG in CTDE and CTCE architectures. The DiaCoG framework is adapted into two variants: DiaCoG-DE for CTDE (indicated by red arrows) and DiaCoG-CE for CTCE (indicated by purple arrows). In the CTDE scheme, the critic network estimates the state value using the DiaCoG representations ($\mathbf{Z}_t$), while actor networks generate actions based on agents' raw observations ($o_t$). The CTCE scheme utilizes DiaCoG representations ($\mathbf{Z}_t$) for action generation by actor networks and for state value estimation by the critic network. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

agents ($\mathbf{Z}$) integrate enriched coordination information, which is derived from agents with strong connections (based on the learned DiaCoG structures), thereby strengthening collaborative decision-making. Specifically, during the training stage, the policy function and the value function are jointly optimized using the objective functions described in Section 2.1. In the execution stage, the trained policy function generates actions based on its inputs, while the value function is not used.

***CTDE.*** In the CTDE architecture, communication between agents is permitted during centralized training but prohibited during decentralized execution. This allows using other agents' information during training, while generating action exclusively depending on each agent's local observations as input during execution. Following [22], the DiaCoG-DE (illustrated by red arrows in Fig. 2) employs the DiaCoG embeddings $\mathbf{Z}_t$ as input to the centralized critic network $\hat{V}_\xi$ to estimate the overall state value, which is further used to calculate the advantage $\hat{A}_t$. Concurrently, the actor network $\pi_{i,\theta}$ generates actions from the raw observations $o_t$, ensuring compatibility with the decentralized execution process.

***CTCE.*** In the CTCE architecture (illustrated by purple arrows in Fig. 2), communication between agents is allowed during both training and execution stages. This enables each agent to utilize information of other agents at both stages. Therefore, DiaCoG-CE directly utilizes embeddings of DiaCoG, $\mathbf{Z}_t$, as input to the policy network $\pi_{i,\theta}$ to generate actions, and as input to the value network $\hat{V}_\xi$ for state value estimation.

## 3. Theoretical analysis

In this section, we show the enhanced expressiveness of representations generated by DiaCoG over those of consistency-based methods, as quantified through information-theoretic measures. Through two theorems, we demonstrate how DiaCoG improves value estimation and action selection in the CTDE and CTCE architectures, respectively. These components—the value estimation and action selection modules—are fundamental to cooperative MARL, enabling agents to coordinate effectively and make collective decisions.

### 3.1. Expressiveness of DiaCoG for value estimation in CTDE architecture

The expressiveness of representations for value estimation is critical to cooperative MARL, as it determines how accurately agents can evaluate joint states to achieve coordinated outcomes. Empirical studies, such as Yu et al. [34], demonstrated that the choice of input to the value function significantly impacts MARL performance in CTDE settings, motivating our theoretical analysis of DiaCoG's representations.

We hypothesize that DiaCoG's representations, which integrate both consistency and discrepancy of agents' observations, enable richer modeling of coordination dynamics compared to representations that rely solely on the consistency/similarity information. This subsection analyzes this advantage theoretically, focusing on how DiaCoG's diametric approach enhances value estimation.

Consider a cooperative MARL environment with $n$ agents at time step $t$ under the CTDE setting. Each agent $i$ receives a local observation $o_{i,t}$, forming the observation set $\boldsymbol{o}_t = \{o_{i,t} \mid i = 1, \dots, n\}$. Let $C_{i,t}$ and $D_{i,t}$ denote the consistency and discrepancy information for agent $i$, respectively, derived from $\boldsymbol{o}_t$ (e.g., via pairwise feature correlations). DiaCoG learns representations $\boldsymbol{z}_{i,t}^{\text{DiaCoG}} = g_\psi(C_{i,t} \oplus D_{i,t})$, where $g_\psi$ is a deterministic, learnable function and $\oplus$ combines inputs, while methods consider solely the consistency information learn $\boldsymbol{z}_{i,t}^{\text{Cons}} = g_\psi(C_{i,t})$. For value estimation, a learning model (usually a neural network) $V_\phi$ estimates the value of the state. The joint state value is computed by aggregating individual values: $\hat{v}_t^{\text{DiaCoG}} = F_\omega\left(\hat{v}_{1,t}^{\text{DiaCoG}}, \dots, \hat{v}_{n,t}^{\text{DiaCoG}}\right)$ for DiaCoG, where $F_\omega$ can be a linear or non-linear function and $\hat{v}_{i,t}^{\text{DiaCoG}} = V_\phi(\boldsymbol{z}_{i,t}^{\text{DiaCoG}})$. Accordingly, for consistency-based methods, $\hat{v}_t^{\text{Cons}} = F_\omega\left(\hat{v}_{1,t}^{\text{Cons}}, \dots, \hat{v}_{n,t}^{\text{Cons}}\right)$, where $\hat{v}_{i,t}^{\text{Cons}} = V_\phi(\boldsymbol{z}_{i,t}^{\text{Cons}})$. The following theorem demonstrates that DiaCoG's representations yield superior expressiveness for value estimation.

**Theorem 1.** *Let $g_\psi$ be a deterministic, learnable function for agent representations, $V_\phi$ a value function, and $C_{i,t}, D_{i,t}$ the consistency and discrepancy information for agent $i$ at time $t$, respectively. Define DiaCoG's representations as $z_{i,t}^{DiaCoG} = g_\psi(C_{i,t} \oplus D_{i,t})$ and consistency-based representations as $z_{i,t}^{Cons} = g_\psi(C_{i,t})$, with estimated values $\hat{v}_{i,t}^{DiaCoG} = V_\phi(z_{i,t}^{DiaCoG})$ and $\hat{v}_{i,t}^{Cons} = V_\phi(z_{i,t}^{Cons})$. Then, the following inequality holds:*

$$I(v_{i,t}^{target}; \hat{v}_{i,t}^{DiaCoG}) \geq I(v_{i,t}^{target}; \hat{v}_{i,t}^{Cons}) \quad \forall i, \tag{8}$$

*where $I(v_{i,t}^{target}; \hat{v}_{i,t}^{DiaCoG})$ denotes the mutual information between the target value $v_{i,t}^{target}$ and the estimated value $\hat{v}_{i,t}^{DiaCoG}$, and $I(v_{i,t}^{target}; \hat{v}_{i,t}^{Cons})$ denotes the mutual information between $v_{i,t}^{target}$ and $\hat{v}_{i,t}^{Cons}$.*

**Proof.** We aim to show that DiaCoG's representations, by incorporating both consistency and discrepancy, yield higher mutual information with the target value $v_{i,t}^{target}$. Define the composite function $f_\theta = V_\phi \circ g_\psi$. For each agent $i$, the estimated value is $\hat{v}_{i,t}^{\text{DiaCoG}} = f_\theta(C_{i,t} \oplus D_{i,t})$ for DiaCoG and $\hat{v}_{i,t}^{\text{Cons}} = f_\theta(C_{i,t})$ for consistency-based methods.

After applying the chain rule of mutual information [36], we have:

$$\begin{aligned}
& I(v_{i,t}^{target}; f_\theta(C_{i,t}), f_\theta(C_{i,t} \oplus D_{i,t})) \\
= \;& H(v_{i,t}^{target}) - H(v_{i,t}^{target} \mid f_\theta(C_{i,t}), f_\theta(C_{i,t} \oplus D_{i,t})) \\
= \;& H(v_{i,t}^{target}) - H(v_{i,t}^{target} \mid f_\theta(C_{i,t})) + H(v_{i,t}^{target} \mid f_\theta(C_{i,t})) - H(v_{i,t}^{target} \mid f_\theta(C_{i,t}), f_\theta(C_{i,t} \oplus D_{i,t})) \\
= \;& I(v_{i,t}^{target}; f_\theta(C_{i,t})) + I(v_{i,t}^{target}; f_\theta(C_{i,t} \oplus D_{i,t}) \mid f_\theta(C_{i,t})).
\end{aligned} \tag{9}$$

Moreover, note that $I(v_{i,t}^{target}; f_\theta(C_{i,t}), f_\theta(C_{i,t} \oplus D_{i,t})) = I(v_{i,t}^{target}; f_\theta(C_{i,t} \oplus D_{i,t}))$, since $f_\theta(C_{i,t})$ is determined by $f_\theta(C_{i,t} \oplus D_{i,t})$. This holds because $C_{i,t} \oplus D_{i,t}$ includes $C_{i,t}$, and $f_\theta$ is deterministic, so knowing $f_\theta(C_{i,t} \oplus D_{i,t})$ implies $f_\theta(C_{i,t})$ with no additional uncertainty. Thus, we have:

$$I(v_{i,t}^{target}; f_\theta(C_{i,t} \oplus D_{i,t})) = I(v_{i,t}^{target}; f_\theta(C_{i,t})) + I(v_{i,t}^{target}; f_\theta(C_{i,t} \oplus D_{i,t}) \mid f_\theta(C_{i,t})). \tag{10}$$

The conditional term, $I(v_{i,t}^{target}; f_\theta(C_{i,t} \oplus D_{i,t}) \mid f_\theta(C_{i,t}))$, is non-negative, representing the additional information about $v_{i,t}^{target}$ provided by $f_\theta(C_{i,t} \oplus D_{i,t})$ given $f_\theta(C_{i,t})$. Since $C_{i,t} \oplus D_{i,t}$ includes $D_{i,t}$, which captures discrepancy features (e.g., distinct agent roles) beyond $C_{i,t}$, this term is generally positive unless $D_{i,t}$ is entirely redundant. Thus, we have:

$$I(v_{i,t}^{target}; f_\theta(C_{i,t} \oplus D_{i,t})) \geq I(v_{i,t}^{target}; f_\theta(C_{i,t})) \quad \forall i, \tag{11}$$

demonstrating that DiaCoG's representations are at least as expressive as consistency-based ones, with equality only if $D_{i,t}$ adds no new information. □

This theorem indicates that DiaCoG's representations, by leveraging both consistency and discrepancy, provide a more informative estimate of the target state value in cooperative MARL with the CTDE architecture. Higher mutual information implies that DiaCoG captures richer dependencies among agents, enabling more accurate value estimation. In CTDE MARL, this translates to better coordination, as agents can align their actions based on a more precise understanding of the joint state, thus improving collective performance.

### 3.2. Coordination effectiveness of DiaCoG for action selection in CTCE architecture

To obtain high collective rewards, the quality of action selection in cooperative MARL is vital for ensuring seamless agent coordination. We hypothesize that DiaCoG's policies, which integrate both consistency and discrepancy information from agents' observations, enable more effective coordination compared to policies that rely solely on consistency information. This subsection investigates this advantage, focusing on how DiaCoG's diametric approach enhances action selection.

Following the cooperative MARL environment with $n$ agents at time step $t$ as defined in Section 3.1, each agent $i$ receives a local observation $o_{i,t}$, forming the observation set $\boldsymbol{o}_t = \{o_{i,t} \mid i = 1, \dots, n\}$. Using the consistency and discrepancy information $C_{i,t}$ and $D_{i,t}$ as defined in Section 3.1, DiaCoG learns representations $\boldsymbol{z}_{i,t}^{\mathrm{DiaCoG}} = g_\psi(C_{i,t} \oplus D_{i,t})$, while methods considering solely the consistency information learn $\boldsymbol{z}_{i,t}^{\mathrm{Cons}} = g_\psi(C_{i,t})$. Each agent $i$ employs a stochastic policy $\pi_{i,\theta}$ with shared parameters $\theta$ to select actions $a_{i,t}$ by inputting the agent features $\boldsymbol{z}_{i,t}$. The following theorem establishes that DiaCoG's policy yields superior effectiveness for action selection.

**Theorem 2.** *Let $g_\psi$ be a learnable function for agent representations, $\pi_{i,\theta}$ a stochastic action policy for agent $i$ with shared parameters $\theta$, and $C_{i,t}$, $D_{i,t}$ the consistency and discrepancy information for agent $i$ at time $t$, respectively. Define DiaCoG's representations as $\boldsymbol{z}_{i,t}^{DiaCoG} = g_\psi(C_{i,t} \oplus D_{i,t})$ and consistency-based representations as $\boldsymbol{z}_{i,t}^{Cons} = g_\psi(C_{i,t})$. Assuming $\pi_{i,\theta}$ is a well-trained policy function that generates action distribution maximizing the expected discounted reward (via surrogate objective that maximizes advantage values) based on input $\boldsymbol{z}_{i,t}$, and $g_\psi$ is deterministic and sufficiently expressive, for the optimal joint action $\boldsymbol{a}_t^*$, the following inequality holds:*

$$p(\boldsymbol{a}_t^* \mid \mathbf{Z}_t^{DiaCoG}) \geq p(\boldsymbol{a}_t^* \mid \mathbf{Z}_t^{Cons}), \tag{12}$$

*where $\mathbf{Z}_t = [\boldsymbol{z}_{1,t}^\top, \dots, \boldsymbol{z}_{n,t}^\top]$ represents the aggregated representations of all agents.*

**Proof.** We aim to show that the embeddings generated by DiaCoG, which incorporates both consistency and discrepancy information, yield a higher probability of selecting the optimal joint action for global coordination. In the following, we prove that the policy $\pi_{i,\theta}$ could leverage the enriched information embedded in the discrepancy information (e.g., differences in agent positions or roles) to select actions that better coordinate across agents.

The non-negativity of mutual information naturally implies the following:

$$I(a_{i,t}^*; \boldsymbol{z}_{i,t}^{\mathrm{DiaCoG}} | \boldsymbol{z}_{i,t}^{\mathrm{Cons}}) \geq 0. \tag{13}$$

Define $\boldsymbol{z}_{i,t}^{\mathrm{DiaCoG}} = g_\psi(C_{i,t} \oplus D_{i,t})$ and $\boldsymbol{z}_{i,t}^{\mathrm{Cons}} = g_\psi(C_{i,t})$, where $g_\psi$ is a deterministic and sufficiently expressive function capable of encoding the valid information from features. Then $\boldsymbol{z}_{i,t}^{\mathrm{Cons}}$ represents a subset of the information contained in $\boldsymbol{z}_{i,t}^{\mathrm{DiaCoG}}$ (recoverable from $C_{i,t} \subset C_{i,t} \oplus D_{i,t}$ via $g_\psi$). As a result, the information about the optimal action $a_{i,t}^*$ derived from $\boldsymbol{z}_{i,t}^{\mathrm{DiaCoG}}$ is at least as comprehensive as that from $\boldsymbol{z}_{i,t}^{\mathrm{Cons}}$, indicating that $D_{i,t}$ may contribute additional insight. This allows us to equivalently rewrite Eq. (13) as $I(a_{i,t}^*; g_\psi(C_{i,t} \oplus D_{i,t}) | g_\psi(C_{i,t})) \geq 0$, with equality holding when $D_{i,t}$ contributes no added value. This suggests that the selection of the optimal action $a_{i,t}^*$ is informed by additional discrepancy information beyond consistency.

Expressing the mutual information in Eq. (13) in terms of the entropy $H(\cdot)$ and the conditional entropy $H(\cdot|\cdot)$, we obtain:

$$\begin{aligned} I(a_{i,t}^*; \boldsymbol{z}_{i,t}^{\mathrm{DiaCoG}} | \boldsymbol{z}_{i,t}^{\mathrm{Cons}}) &= H(a_{i,t}^* \mid \boldsymbol{z}_{i,t}^{\mathrm{Cons}}) - H(a_{i,t}^* \mid \boldsymbol{z}_{i,t}^{\mathrm{Cons}}, \boldsymbol{z}_{i,t}^{\mathrm{DiaCoG}}), \\ &= H(a_{i,t}^* \mid \boldsymbol{z}_{i,t}^{\mathrm{Cons}}) - H(a_{i,t}^* \mid \boldsymbol{z}_{i,t}^{\mathrm{DiaCoG}}) \geq 0, \end{aligned} \tag{14}$$

which leads to:

$$H(a_{i,t}^* | \boldsymbol{z}_{i,t}^{\mathrm{DiaCoG}}) \leq H(a_{i,t}^* | \boldsymbol{z}_{i,t}^{\mathrm{Cons}}). \tag{15}$$

Since the MAPPO surrogate objective maximizes expected advantage, it encourages probability mass shift toward actions with positive advantage, including $a_{i,t}^*$. The lower conditional entropy on the left-hand side of Eq. (15) implies that $\boldsymbol{z}_{i,t}^{\mathrm{DiaCoG}}$ provides more certainty about $a_{i,t}^*$. In a DEC-POMDP setting, access to richer information would typically facilitate the selection of more effective actions for the global coordination. The increased certainty likely enables the policy to assign a higher probability to $a_{i,t}^*$:

$$\pi_{i,\theta}(a_{i,t}^* \mid \boldsymbol{z}_{i,t}^{\mathrm{DiaCoG}}) \geq \pi_{i,\theta}(a_{i,t}^* \mid \boldsymbol{z}_{i,t}^{\mathrm{Cons}}) \quad \forall i, \tag{16}$$

indicating that the probability of generating the optimal action for each agent based on DiaCoG is at least as high as that of the consistency-based approaches. Since actions are sampled independently from each agent's policy conditioned on its respective representation, the joint probability of the optimal joint action $\boldsymbol{a}_t^*$ is the product over all agents. Thus, we have:

$$p(\boldsymbol{a}_t^* \mid \mathbf{Z}_t^{\mathrm{DiaCoG}}) = \prod_{j=1}^{n} \pi_{j,\theta}(a_{j,t}^* \mid \boldsymbol{z}_{j,t}^{\mathrm{DiaCoG}}) \geq \prod_{j=1}^{n} \pi_{j,\theta}(a_{j,t}^* \mid \boldsymbol{z}_{j,t}^{\mathrm{Cons}}) = p(\boldsymbol{a}_t^* \mid \mathbf{Z}_t^{\mathrm{Cons}}), \tag{17}$$

where $\mathbf{Z}_t = [\boldsymbol{z}_{1,t}^\top, \dots, \boldsymbol{z}_{n,t}^\top]$ denotes the aggregated representations of all agents. This indicates that the probability of generating the optimal joint action based on DiaCoG embeddings is at least as high as that of consistency-based approaches. □

This theorem suggests that DiaCoG's action selection, by combining consistency and discrepancy information, enhances the quality of actions within the CTCE architecture in cooperative MARL. By uncovering deeper inter-agent relationships and integrating diverse perspectives, DiaCoG supports action selection that aligns with the optimal action, thus leading to improved coordination in the CTCE setting.

## 4. Experimental framework and environment overview

### 4.1. Overview of experimental environments

We conducted a systematic evaluation of the proposed DiaCoG framework by comparing its performance with several baseline and state-of-the-art (SOTA) algorithms across three representative cooperative MARL environments: Predator-Prey [11], Traffic Junction [17], and StarCraft II Multi-Agent Challenge (SMAC) [37]. These three environments are widely recognized benchmarks that necessitate effective coordination among agents to achieve collective goals, making them appropriate for assessing DiaCoG's capabilities. In the Predator-Prey environment, predators must collaborate to capture prey within a grid-based space. This task requires the predators to learn and develop behaviors of coordination, pursuit, and trapping. The penalty for single-agent capture attempts can be adjusted to incentivize collaborative strategies. The Traffic Junction environment simulates autonomous vehicle interactions, requiring agent coordination to navigate roads safely, avoid collisions, cross intersections, and prevent congestion. The SMAC environment involves micro-management tasks where allied units must strategize to maximize damage against enemy forces and secure victory across diverse map settings, necessitating teamwork and tactical planning. Detailed configurations and specific settings for these environments are provided in Sections 5.1, 7.1, and 8.1, respectively.

### 4.2. Baselines and implementation details

In the classic Predator-Prey and Traffic Junction environments, we evaluated the variants of our proposed DiaCoG framework, DiaCoG-CE and DiaCoG-DE (i.e., DiaCoG with the centralized and decentralized execution architectures) by comparing them with the following algorithms: (1) Multi-Agent PPO with Centralized Training and Centralized Execution (denoted as MAPPO-CE), which employs a centralized actor and a centralized critic that both take the concatenated observations of all agents as input; (2) Multi-Agent PPO with Centralized Training and Decentralized Execution (denoted as MAPPO-DE), which uses a decentralized actor taking local observations as input and a centralized critic taking the concatenated observations of all agents as input; (3) Deep Implicit Coordination Graphs (DICG)[2] [22], which utilizes the self-attention mechanism to infer implicit coordination structures, implemented in a CTCE framework (denoted as DICG-CE); (4) DICG implemented in a CTDE framework (denoted as DICG-DE); (5) Heterogeneous Agent Trust Region Policy Optimization (HATRPO)[3] [27,29], which is a SOTA method incorporating heterogeneous policy networks and a sequential update scheme into TRPO; and (6) Heterogeneous Agent Proximal Policy Optimization (HAPPO), which incorporates heterogeneity into MAPPO with a scheme similar to that in HATRPO.

The SMAC environment has emerged as a prominent benchmark for cooperative MARL algorithms due to its diverse, challenging maps and standardized testbeds. To demonstrate the adaptability of DiaCoG across different MARL paradigms, we integrated it with a SOTA value-based approach, the Group-Aware Coordination Graph (GACG) [15]. To comprehensively evaluate the performance of our algorithm in the SMAC environment, we compare our DiaCoG-enhanced variant, GACG-DiaCoG, with several representative and SOTA algorithms, including the original GACG[4] [15], QMIX [13], Deep Coordination Graphs (DCG) [11], and Latent Temporal Sparse Cooperative Graph (LTS-CG)[5] [16]. Among them, QMIX and DCG serve as leading representatives of value decomposition and coordination graph methods, respectively, while GACG and LTS-CG represent more recent advances in coordination graph-based methods: GACG employs attention mechanisms to learn agent-pair structures, and LTS-CG learns latent temporal sparse graphs to capture inter-agent dependencies.

For implementation, we utilized the original source codes of the comparative algorithms, adhering to their provided settings to ensure a fair evaluation. For MAPPO-CE, MAPPO-DE, DICG-CE, and DICG-DE, as well as our approaches DiaCoG-DE and DiaCoG-CE, the policy function is implemented using a multilayer perceptron (MLP) network; for the GACG and GACG-DiaCoG, we adopted identical model architectures, except for the DiaCoG construction module, thereby ensuring a consistent architecture for comparison. All reported results are averaged over 5 random seeds, enhancing the reliability of the experimental results.

## 5. Predator-Prey

### 5.1. Environment description and settings

The Predator-Prey environment used in our experiments is a grid-world environment with discrete action and state spaces, also adopted by Böhmer et al. [11] and Li et al. [22]. This environment consists of a $k \times k$ grid populated by $n$ predator agents and $m$ prey, as illustrated in Fig. 3, where $k = n = m = 12$, and blue nodes represent predators while red nodes denote prey.

The objective of the predator agents, whose policies need to be learned, is to capture the prey. Both predators and prey can move within this grid world. For a successful capture, it requires at least two predators to simultaneously occupy grid cells adjacent to a prey (e.g., in Fig. 3, predators 1 and 8 successfully capture prey 1, and predators 2 and 7 successfully capture prey 3). Upon successful capture, the prey will be removed from the current environment. An episode terminates when either (1) the maximum number of episode steps (200 in our experiments) is reached, or (2) all prey have been removed. A reward of 10 is awarded to each successful

[2] https://github.com/sisl/DICG. Accessed February 9, 2026
[3] https://github.com/morning9393/HAPPO-HATRPO. Accessed March 19, 2026
[4] https://github.com/Wei9711/GACG. Accessed February 9, 2026
[5] https://github.com/Wei9711/LTSCG. Accessed February 9, 2026

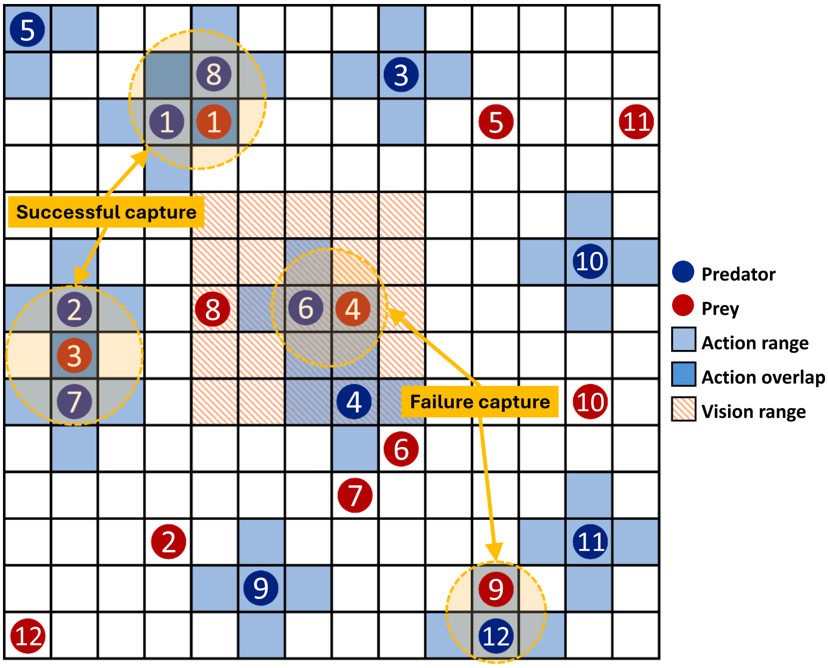



**Fig. 3.** Predator-Prey environment with $12 \times 12$ grids, 12 predators (indicated by the blue circles) and 12 prey (indicated by the red circles). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

capture. If a single predator attempts to capture the prey solely, there will be a negative reward of $p$ points as the punishment (e.g., predator6′s failed attempt on prey 4 and predator12′s failed attempt on prey 9, as shown in Fig. 3). Additionally, each predator incurs a step cost of $-0.2$. From the above settings, we can observe that these reward and penalty mechanisms encourage cooperation between agents.

The action space of each predator is discrete and consists of five actions: move up, move down, move left, move right, and stay, as indicated by the blue action range of each predator in Fig. 3. These actions determine the predator's movement within the grid. The observation for each predator can be represented as a vector, which comprises: (1) the predator's current coordinates, (2) a $5 \times 5$ local view centered on the predator's position (e.g., the red vision range around predator 6 in Fig. 3), indicating the presence of prey in each cell, (3) the ratio of current step count to the maximum episode steps, and (4) a $5 \times 5$ local view centered on the predator's position, indicating the presence of other predators in each cell.

We experiment with penalty values $p \in \{0, -1, -1.25, -1.5\}$. All comparison algorithms were trained for 1000 epochs in the $p = 0$ scenario and trained for 1500 epochs in the remaining three scenarios, corresponding to approximately 5 million and 7.5 million environment steps, respectively.

### 5.2. Experimental results and discussion

For the Predator-Prey environment, Fig. 4 shows the evolution of the evaluated average returns, tested over 100 evaluation episodes per 5 epoch intervals, for various algorithms under four scenarios with different penalty values. Each subfigure on the left-hand side corresponds to a specific penalty value $(0, -1, -1.25, -1.5)$, while the zoomed-in plot on the right highlights local convergence trends. To mitigate observed oscillations approaching convergence caused by the randomness of test environments, we report the mean and standard deviation of the evaluated average return averaged over the last 15 tests, computed across five random seeds. Numerical results are provided in Table 1.

Overall, our methods, DiaCoG-CE-MLP and DiaCoG-DE-MLP, consistently outperform the baseline approaches in most of the scenarios, in terms of convergence speed and final return values, validating DiaCoG's effectiveness in enhancing coordination. Specifically, in the scenario with no penalty ($penalty = 0$), the final return values show small variation across algorithms, reflecting a relatively straightforward setting where most approaches converge rapidly. The zoomed-in plot on the right-hand side of the first subfigure in Fig. 4 reveals that centralized architectures, such as DICG-CE and DiaCoG-CE, generally achieve higher final return values, except for MAPPO-CE-MLP, which slightly struggles in the scenario with no penalty and deteriorates in other three scenarios with increasing penalty due to the large joint observation space and inefficient information extraction, as noted in [22]. At the meantime, the HATRPO and HAPPO demonstrate competitive performance in this scenario.

For penalty values of $-1.0$, $-1.25$, and $-1.5$, which increasingly penalize single-agent capture attempts and thus encourage cooperation, the performance distinctions become more obvious. In general, coordination graph-based methods (DICG and DiaCoG variants) outperform methods lacking explicit coordination mechanisms, i.e., MAPPO-DE-MLP, and methods incorporating the coordination

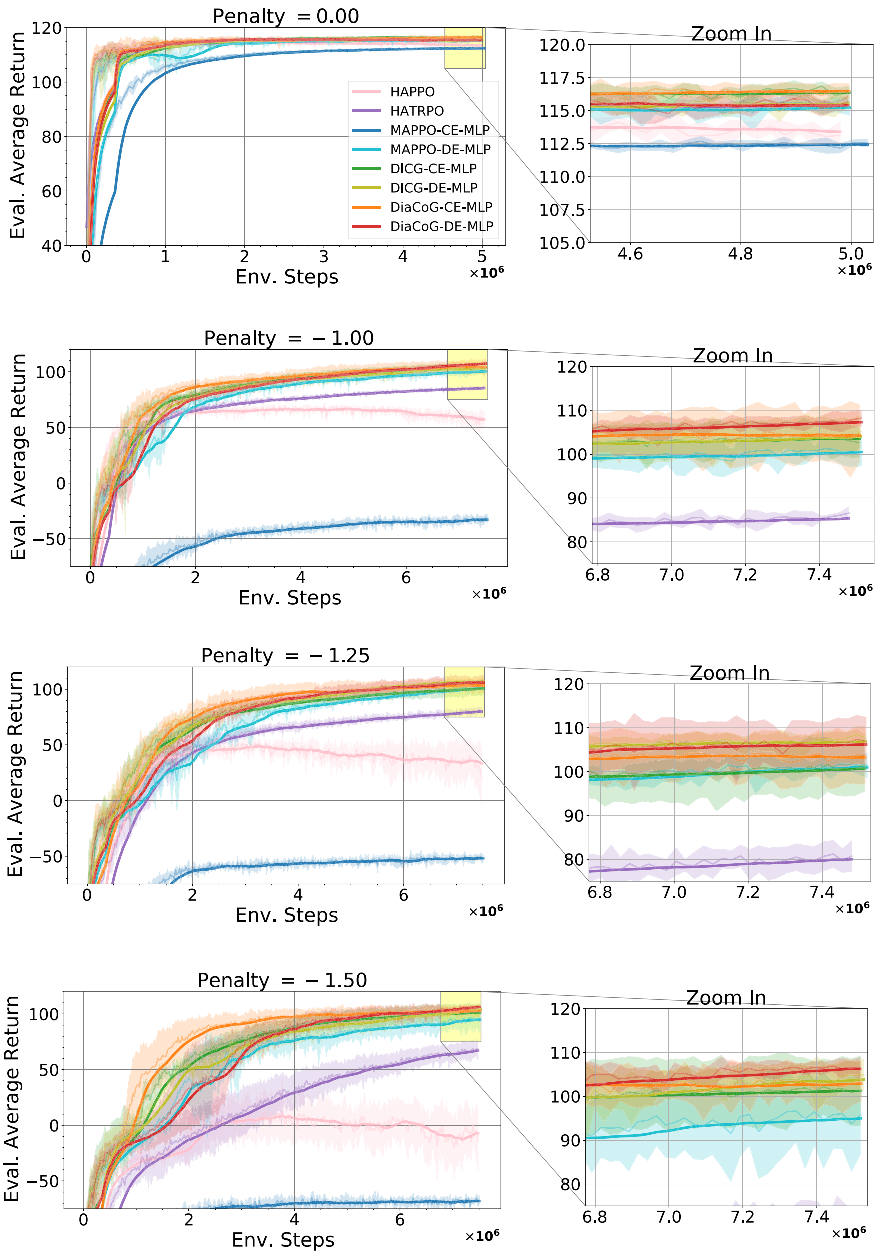



**Fig. 4.** Learning curves of average returns in the Predator-Prey environment under different penalty values $(0, -1, -1.25, -1.5)$. Each subfigure shows the mean (semitransparent solid line) and standard deviation (shaded area) of evaluated average returns over 5 runs. The opaque solid line represents the average smoothed over the 15 test intervals. Zoomed-in plots on the right highlight local convergence trends.

with heterogeneous policy networks and sequential update scheme, i.e., HATRPO and HAPPO. Furthermore, as presented in Fig. 4 and Table 1, DiaCoG variants show slightly but consistently higher evaluated average returns over DICG counterparts in most cases when they both approach near-optimum. It empirically confirms that the diametric coordination graph can enhance policy learning by leveraging both consistency and discrepancy information.

## 6. Case study on Predator-Prey

In this section, we present a case study in the Predator-Prey environment to examine how the proposed DiaCoG fosters effective coordination, illustrating its advantages over existing consistency-based methods.

**Table 1**
Evaluated average returns (mean and standard deviation computed across five random seeds), averaged over the last 15 tests in the Predator-Prey environment across different penalties. The best results for each penalty value are highlighted in boldface.

| | 0.00 | −1.00 | −1.25 | −1.50 |
|---|---|---|---|---|
| HAPPO | 113.39 (0.50) | 57.19 (6.00) | 33.17 (17.63) | −7.07 (19.89) |
| HATRPO | 115.37 (0.06) | 85.37 (0.66) | 80.01 (2.72) | 67.02 (7.55) |
| MAPPO-CE-MLP | 112.43 (0.14) | −32.88 (2.76) | −51.80 (2.46) | −68.11 (1.50) |
| MAPPO-DE-MLP | 115.24 (0.21) | 100.48 (2.79) | 100.96 (2.06) | 94.97 (9.14) |
| DICG-CE-MLP | 116.38 (0.32) | 103.51 (3.69) | 100.68 (6.22) | 101.23 (7.42) |
| DICG-DE-MLP | 115.50 (0.27) | 104.05 (4.12) | **106.84 (1.59)** | 103.83 (3.37) |
| DiaCoG-CE-MLP | **116.47 (0.50)** | 104.24 (6.53) | 103.22 (5.73) | 102.89 (4.57) |
| DiaCoG-DE-MLP | 115.41 (0.31) | **107.29 (1.55)** | 106.16 (5.34) | **106.28 (1.43)** |

### 6.1. Impact of agent features on coordination weights

We first analyze how the integration of consistency and discrepancy information influences the connection weights in the learned coordination graph. Fig. 5 illustrates the relationship between the cosine similarity of agent observation embeddings and the normalized edge weights of the diametric coordination graph **G** across various penalty settings. Specifically, each subfigure compares the weight patterns of DICG and DiaCoG under the centralized architecture for penalty values of 0, −1, −1.25, and −1.5, respectively. At a penalty of 0, where no punishment is imposed for the uncooperative behavior (though failed captures still occur), DiaCoG assigns higher weights to both the agents with consistent observations and those with discrepant observations, aligning with its design to leverage both **C** and **D**. In contrast, DICG mainly prioritizes agents with higher cosine similarity when no penalty ($p = 0$) or a small penalty ($p = -1$) is induced. As the penalty for the uncooperative behavior increases (e.g., −1.25 and −1.5), both methods assign higher weights to agents with discrepant observations, and this pattern is consistently more evident in DiaCoG than in DICG.

Notably, DICG, initially dominated by similarity at zero or small penalties, adapts to give higher weights to agents with discrepant observations as penalties rise. This adaptation reflects an environment-driven response to encourage cooperation and suggests that increasing penalties incentivize more complex cooperative strategies, such as joint prey capture. However, DiaCoG's mechanism inherently takes discrepancy into account from the outset, even without penalties, demonstrating its robustness. This intrinsic design avoids relying on external settings to encourage collaboration, which in practice may introduce extra costs or cause overly cautious behavior (e.g., prolonged capture times). This underscores DiaCoG's advantage in flexibly promoting collaboration across diverse observation profiles.

### 6.2. Collaboration between agents with divergent observations

In this subsection, we illustrate how DiaCoG promotes enhanced coordinated behavior through a specific case of two close steps (step 9 and step 11) in the Predator-Prey environment, conducted under a penalty of 0. As shown in Fig. 6, at step 9 (the left subfigure), for predator 5, predator 1 exhibits the highest cosine similarity in terms of the observation embeddings, largely due to their close positions and overlapping observation fields. However, DiaCoG assigns higher weights to predators 4 and 3 for predator 5, despite its lower cosine similarity and greater distance from predator 5, as shown by the weight values on the light black background. Transitioning to step 11, the prey 1 (highlighted by an orange circle in the left subfigure) is moving up, whose direction is indicated by a red arrow. As shown in the right subfigure in Fig. 6, predator 5 moves left and up, and predator 4 moves down, as indicated by their adjusted positions and action directions marked by the orange arrows. These actions result in predator 5 and predator 4 successfully capturing prey 1 (marked as a circle filled by gray in the right subfigure). This case illustrates that DiaCoG's weight assignment extends beyond consistency. By incorporating the discrepancy from agents with divergent observations, agents obtain a supplementary view for decision-making, so as to facilitate a strategic approach to prey capture from different directions.

In contrast, proximity-based methods, which rely on the intuition of close collaboration mainly emerging between agents with similar observations (as shown in Fig. 6), would favor predator 1 due to its higher cosine similarity with predator 5. These approaches, however, could overlook the potential of predators 4, whose divergent perspectives could provide complementary information and actions to enhance capture success. By integrating discrepancy alongside consistency, DiaCoG addresses this limitation inherently, improving the coordination effectiveness through its fundamental mechanism rather than relying solely on the incentive settings of environments.

## 7. Traffic Junction

### 7.1. Environment description and settings

The Traffic Junction environment is also a grid-world multi-agent environment [17,18,22], as shown in Fig. 7. In this environment, agents, represented as cars, are randomly introduced to the road network with pre-assigned grid layouts at a specified probability $p_{arr}$. Their primary goals are to avoid collisions (i.e., two cars occupy the same cell simultaneously) and reach their destinations (i.e.,

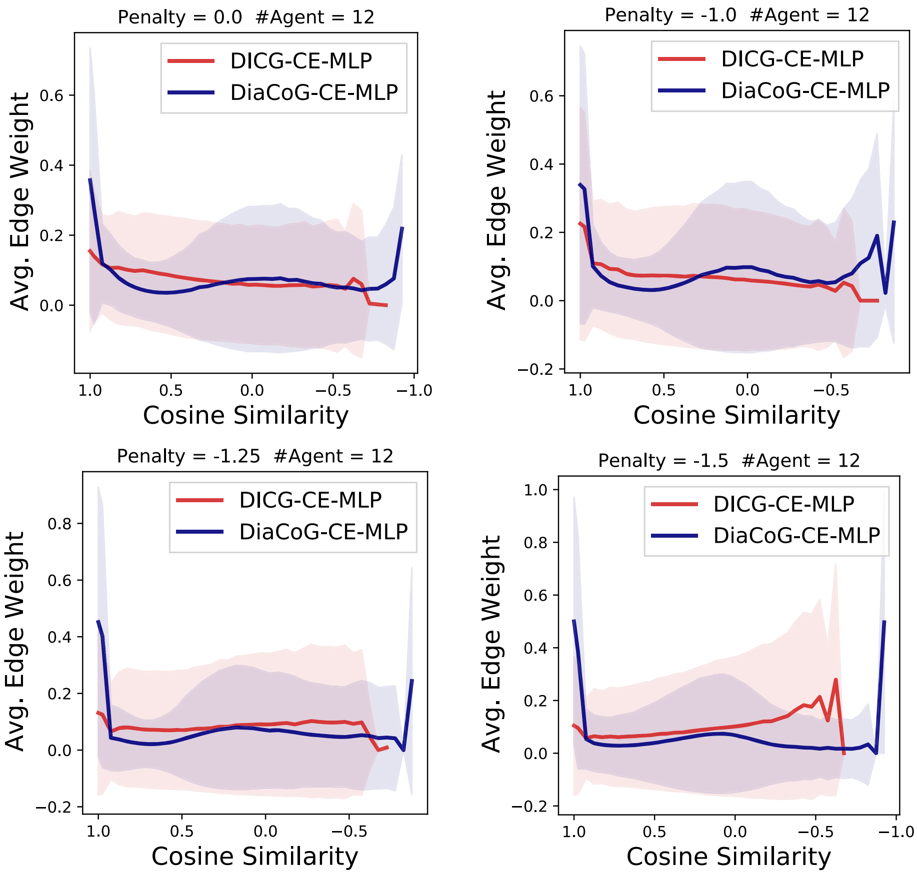



**Fig. 5.** Relationship between cosine similarity of agent observation embeddings and average edge weights in the learned coordination graph. Solid lines represent mean values, with shaded areas denoting standard deviation. Each approach was trained with 5 random seeds, and each trained model was evaluated over 100 episodes, computing cosine similarity and weights per step, then averaging with standard deviation.

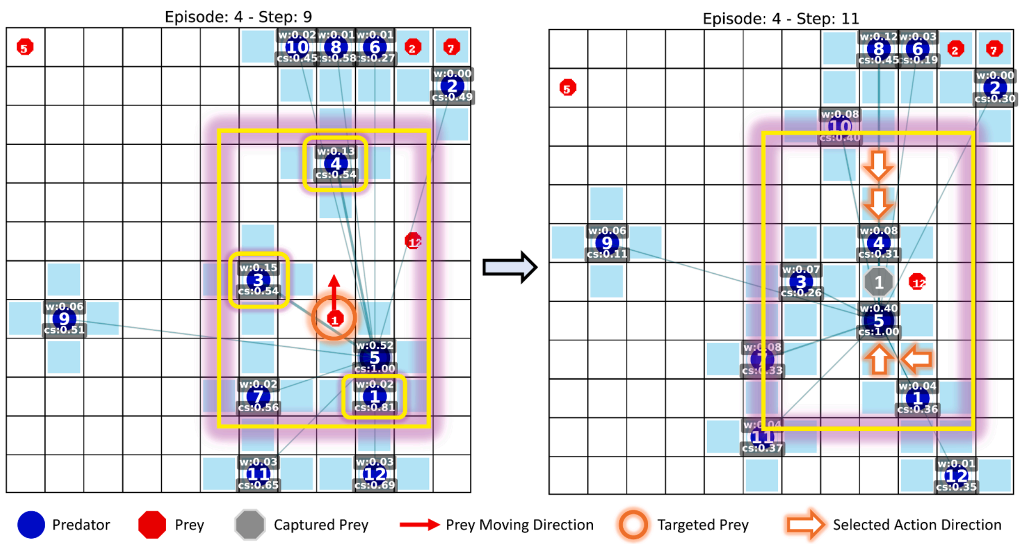



**Fig. 6.** Two close steps (step 9 and step 11) illustrating agent collaboration in the Predator-Prey environment. Blue circles denote predators, and red circles denote prey, with white numbers indicating indices of predators and prey. Light blue shadows show available action directions of each agent, dark green lines represent connections with weights, and the text on light black backgrounds displays weights and cosine similarity values between the agent of interest and other agents. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

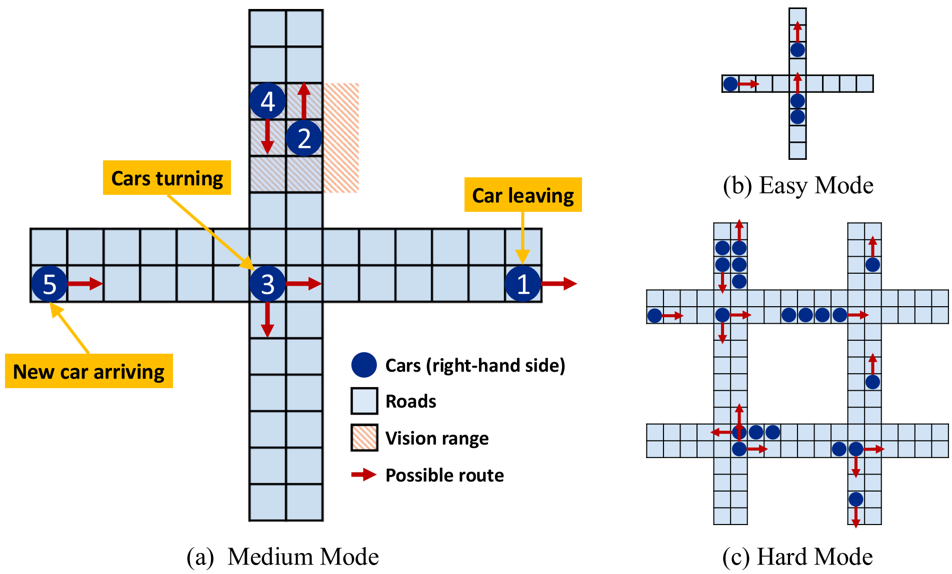



**Fig. 7.** Traffic Junction environment with various difficulty levels: (a) medium mode, (b) easy mode, and (c) hard mode.

edges of the grid). To penalize collisions, a collision penalty (i.e., $r_{coll} = -10$) is applied when two cars collide; to discourage traffic congestion, a step cost proportional to existence time, $r_{time}\tau = -0.01\tau$, is imposed on each car at every time step, where $\tau$ denotes the duration of the car's presence in the grid. The total reward at time step $t$ is thus formulated as:

$$r(t) = C_t r_{coll} + \sum_{i=1}^{n_t} r_{time}\tau_i, \tag{18}$$

where $C_t$ is the number of collisions at time step $t$, and $n_t$ is the number of cars appearing in the grid at time step $t$. The simulation duration of each episode varies with the difficulty level, and the scenario is classified as a failure if collisions occur during the entire episode. Each car has two discrete actions: (1) gas, moving forward one cell along its assigned route; (2) brake, remaining stationary. The movement of cars adheres to the right-hand traffic rule on two-way roads, and at junctions, cars are assigned to one of several possible routes (e.g., turn directions or go straight), as indicated by the red arrows in Fig. 7. The observation for each car encompasses a $3 \times 3$ cell region centered on its position, as illustrated by the orange shadow in Fig. 7. This observation is encoded as a one-hot vector concatenating three types of information: (1) the last action taken, (2) the route ID, and (3) the features of the 9 observation cells.

We adopt the environment settings from [22], which defines three difficulty levels demonstrated in Fig. 7: (1) easy mode, with a maximum of 5 agents per time step, a maximum of 20 environment steps, and a map of 2 one-way roads on a $9 \times 9$ grid; (2) medium mode, with a maximum of 10 agents per time step, a maximum of 40 environment steps, and a map of a junction with 2 two-way roads on a $14 \times 14$ grid; and (3) hard mode, with a maximum of 20 agents per time step, a maximum of 60 environment steps, and a map consisting of 4 connected junctions with 4 two-way roads on an $18 \times 18$ grid. The training comprises 1000 epochs, which induces approximately 16 million, 8 million, and 4 million total environment steps for the easy, medium, and hard scenarios, respectively.

### *7.2. Experimental results and discussion*

For the Traffic Junction environment, Fig. 8 presents the evolution of the tested success rates per 5 training epochs for various algorithms across three difficulty levels, with each subfigure corresponding to a specific difficulty level. Results for the tested success rate (mean and standard deviation computed across five random seeds) averaged over the last 15 tests are reported in Table 2. Overall, the proposed methods consistently demonstrate superior performance compared to the baseline approaches, achieving faster convergence and higher success rates in most scenarios. Specifically, in the easy mode, all methods achieve comparably high success rates, with HATRPO performing the best. As the difficulty escalates to medium and hard modes, the performance gaps between algorithms widen, reflecting the greater need for effective coordination as task complexity increases. In these challenging scenarios, MAPPO-CE-MLP exhibits significant failure, likely attributable to its inefficiency in handling the large joint observation space. In contrast, decentralized approaches, notably our DiaCoG-DE-MLP, outperform their centralized counterparts and achieve the highest success rates with rapid convergence. Meanwhile, HATRPO and HAPPO do not perform well in the medium and hard modes. This is likely because their inherent algorithm design, i.e., training separate policy networks for a fixed maximum number of agents, is not well suited to environments with dynamically changing agent numbers. Table 2 reinforces these observations: our methods remain competitive in the easy mode, while DiaCoG-DE-MLP leads in medium and hard modes. These results provide empirical evidence that DiaCoG's integration of discrepancy information enhances adaptive coordination in increasingly challenging traffic scenarios.

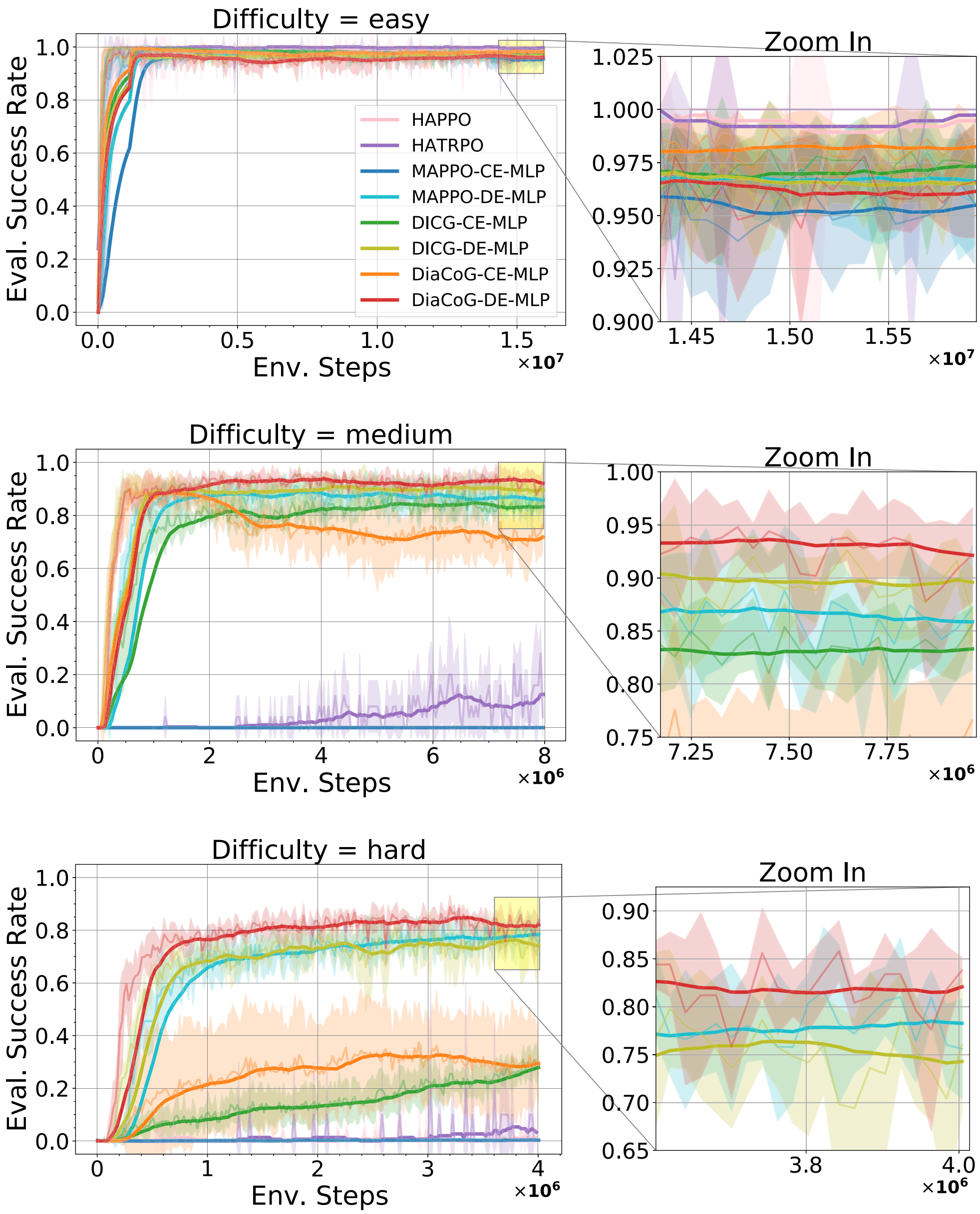



**Fig. 8.** Learning curves of tested success rate in the Traffic Junction environment under different difficulty levels (easy, medium, hard). Each subfigure displays the mean (semitransparent solid line) and standard deviation (the shaded area) of the tested success rate over 5 runs for each algorithm. The opaque solid lines denote the average smoothed over the 15 test intervals. Zoomed-in plots on the right highlight local convergence trends.

## 8. StarCraft II Multi-Agent Challenge (SMAC)

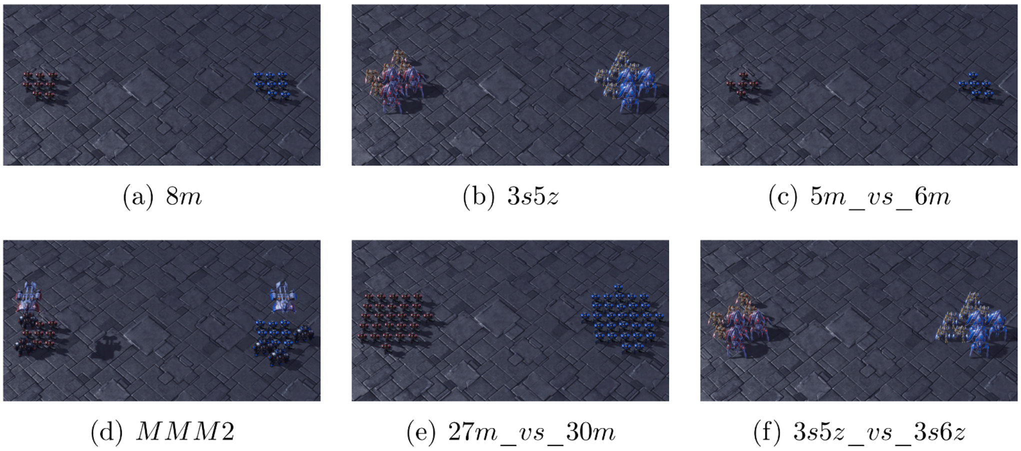
### 8.1. Environment description and settings

The SMAC is a cooperative MARL environment [37] built upon the StarCraft II real-time strategy (RTS) game.[6] It features micromanagement scenarios where each agent controls an allied unit and must coordinate with other allied units to combat enemy units controlled by the built-in game AI with configurable difficulty levels. Detailed specifications are available in the official doc-

[6] https://github.com/oxwhirl/smac/tree/master. Accessed February 9, 2026

**Table 2**
Tested success rate (mean and standard deviation computed across five random seeds), averaged over the last 15 tests in the Traffic Junction environment under three difficulty levels. The best results for each difficulty level are highlighted in boldface.

| | easy | medium | hard |
|---|---|---|---|
| HAPPO | 99.5% (0.7%) | 0.0% (0.0%) | 1.1% (1.0%) |
| HATRPO | **99.7% (0.5%)** | 12.5% (4.0%) | 3.3% (0.0%) |
| MAPPO-CE-MLP | 95.5% (2.1%) | 0.1% (0.0%) | 0.3% (0.1%) |
| MAPPO-DE-MLP | 96.6% (0.5%) | 85.9% (4.6%) | 78.3% (3.1%) |
| DICG-CE-MLP | 97.3% (1.4%) | 83.3% (1.3%) | 27.9% (3.9%) |
| DICG-DE-MLP | 96.5% (0.5%) | 89.6% (1.8%) | 74.3% (5.5%) |
| DiaCoG-CE-MLP | 98.2% (1.1%) | 71.8% (8.2%) | 29.4% (17.9%) |
| DiaCoG-DE-MLP | 96.1% (0.4%) | **92.1% (3.2%)** | **82.1% (2.1%)** |



(a) $8m$ (b) $3s5z$ (c) $5m\_vs\_6m$ (d) $MMM2$ (e) $27m\_vs\_30m$ (f) $3s5z\_vs\_3s6z$

**Fig. 9.** Snapshots of six maps (i.e., $8m$, $3s5z$, $5m\_vs\_6m$, $MMM2$, $27m\_vs\_30m$, and $3s5z\_vs\_3s6z$) of StarCraft II.

umentation[7]. The primary goal is to maximize the win rate, with rewards being designed to reflect combat outcomes. Specifically, a win in an episode yields 200 points, a loss yields 0 points, and killing an enemy unit contributes 10 points, ensuring a non-sparse reward structure. The action space for agents is discrete, comprising move (four directions: north, south, east, or west), attack enemy, stop, and no-op (applicable only to dead agents). For Medivac units, a healer type, the action set includes heal allied units instead of attack. Each agent receives a limited local observation within its sight range, represented as a feature vector consisting of the distance, relative $x-$ and $y-$coordinates, health, shield, and unit type of both allied and enemy units.

In our experiments, we compared our approach, i.e., GACG-DiaCoG, with GACG, QMIX, DCG, and LTS-CG on six maps spanning easy to super-hard difficulties (see Table 3 for details and Fig. 9 for illustrative screenshots). We used StarCraft II game version 5.0.14 on the maps $8m$, $3s5z$, and $MMM2$, and version 4.6.2 on the remaining maps. In all cases, the opponent game AI was set to difficulty level 7 (Very Hard). Each algorithm was trained with 5 random seeds to ensure the reliability of the results. Training configurations differ by map: For the $8m$ map, training involves 0.8 million environment steps with an evaluation/testing interval of 4 thousand steps; for the $3s5z\_vs\_3s6z$ map, training involves approximately 4 million environment steps with a 20-thousand-step interval for evaluation; and for all other maps, training involves approximately 2 million environment steps with a 10-thousand-step interval for evaluation.

### *8.2. Experimental results and discussion*

Fig. 10 presents the learning curves of the evaluated win rate for the DiaCoG-enhanced variant (i.e., GACG-DiaCoG) and the baseline methods across six SMAC maps: $8m$, $3s5z$, $5m\_vs\_6m$, $MMM2$, $27m\_vs\_30m$, and $3s5z\_vs\_3s6z$. Each subfigure corresponds to a specific map, with smoothed averages over 15 test intervals highlighting convergence trends. We observe that GACG-DiaCoG consistently outperforms the original GACG on most maps, with particularly clear advantages in super-hard scenarios ($MMM2$, $27m\_vs\_30m$, and $3s5z\_vs\_3s6z$). In these cases, the proposed GACG-DiaCoG achieves higher final win rates, providing evidence that the diametric coordination module can be integrated effectively into existing coordination graph architectures and yields meaningful gains. Moreover, compared to other baselines (QMIX, DCG, and LTS-CG), GACG-DiaCoG delivers competitive results on easier

[7] https://github.com/oxwhirl/smac/blob/master/docs/smac.md. Accessed February 9, 2026

**Table 3**
Map specifications for SMAC environment. Details include the number and types of allied and enemy units, the homogeneity and symmetry characteristics of each map, and the map difficulty.

| Map Name | Ally Units | Enemy Units | Type | Difficulty |
|---|---|---|---|---|
| $8m$ | 8 Marines | 8 Marines | Homogeneous & Symmetric | easy |
| $3s5z$ | 3 Stalkers & 5 Zealots | 3 Stalkers & 5 Zealots | Heterogeneous & Symmetric | easy |
| $5m\_vs\_6m$ | 5 Marines | 6 Marines | Homogeneous & Asymmetric | hard |
| $MMM2$ | 1 Medivac, 2 Marauders, & 7 Marines | 1 Medivac, 3 Marauders, & 8 Marines | Heterogeneous & Asymmetric | super hard |
| $27m\_vs\_30m$ | 27 Marines | 30 Marines | Homogeneous & Asymmetric | super hard |
| $3s5z\_vs\_3s6z$ | 3 Stalkers & 5 Zealots | 3 Stalkers & 6 Zealots | Heterogeneous & Asymmetric | super hard |

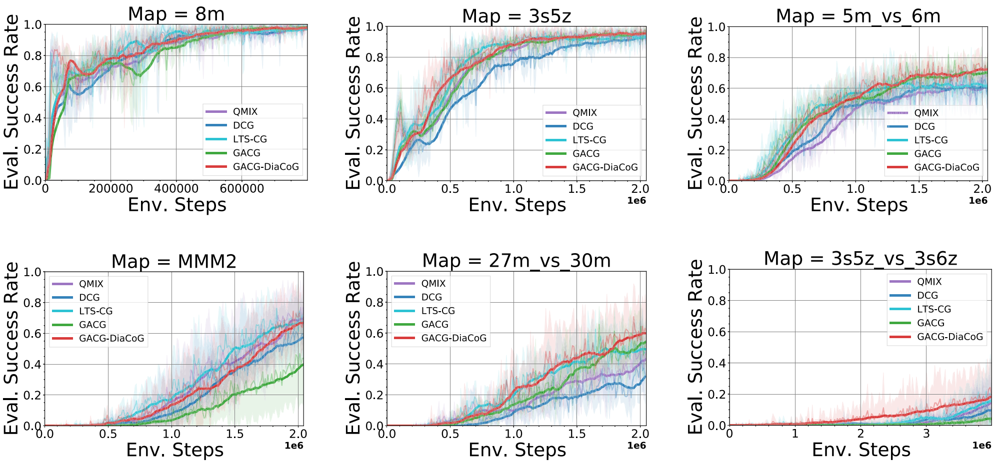



**Fig. 10.** Learning curves of the evaluated win rate in the SMAC environment across different maps. Each subfigure shows the mean win rate (semitransparent solid line) and standard deviation (shaded area) over 5 runs for each algorithm, with smoothed averages over 15 test intervals (opaque solid line) to highlight learning trends across six maps: $8m$, $3s5z$, $5m\_vs\_6m$, $MMM2$, $27m\_vs\_30m$, and $3s5z\_vs\_3s6z$.

maps, such as $8m$ and $3s5z$, while achieving superior performance on $5m\_vs\_6m$, $27m\_vs\_30m$, and $3s5z\_vs\_3s6z$. These improvements in asymmetric and heterogeneous scenarios suggest that incorporating both consistency and discrepancy information fosters more effective cooperative strategies among agents toward the shared objective.

## 9. Related work

Existing research on fostering coordination behaviors in cooperative MARL can be broadly divided into three categories: (1) explicitly or implicitly modeling coordination relationships by decomposing the joint value function; (2) enhancing cooperative behaviors through communication and relation modeling, and (3) designing mechanisms to incorporate behavioral diversity and agent heterogeneity.

### *9.1. Explicitly/Implicitly modeling coordination via decomposing joint value function*

Coordination Graphs (CGs), initially proposed to explicitly model agent collaboration in multi-agent systems, factorize the global Q-function into a linear combination of local Q-functions (i.e., utility and pairwise payoff functions) [10,38]. This strategy effectively mitigates the computational complexity of the joint state-action space, which grows exponentially with the number of agents and actions in MARL. To enhance representation capacity and address relative overgeneralization [39], Deep CGs (DCGs) extend CGs by using parameter-sharing deep neural networks to approximate the pairwise payoff function [11]. However, CGs typically rely on pre-defined graph topologies, often requiring domain-specific prior knowledge, whereas DCGs usually perform best with fully connected graphs [11]. Furthermore, both CGs and DCGs assume static, undirected, and unweighted graph structures, which could limit their practical applicability in complex scenarios. Therefore, recent approaches extend CGs and DCGs to explore dynamic and sparse structures [40].

Another important category of approaches in cooperative MARL involves learning the decomposition of the joint value function without pre-defining the coordination structure. They establish relationships between joint and individual value functions through a set of constraints. Methods like Value-Decomposition Network (VDN) [12] use additive decomposition, while QMIX [13] imposes a monotonicity constraint to enable non-linear decompositions, and QTRAN [14] further relaxes these constraints for greater flexibility. Building on these value decomposition techniques, recent work further integrates graph structure modeling to enhance coordination learning [15,16,41].

### 9.2. Enhancing coordination via communication and relation modeling

Agent communication and relation modeling have become central themes for enhancing the coordination performance in cooperative MARL. One important direction involves explicit communication protocols that enable agents to share information directly. Representative examples include CommNet [17], which enables continuous communication among all agents. Subsequent work has incorporated gate and attention mechanisms to make communication more selective and efficient [18–20,42]. A parallel line of research shifts focus from explicit message passing to inferring implicit inter-agent relationships, often represented as graphs. These methods learn abstract relational structures to support coordination without direct communication channels. Attention-based graph methods for coordination modeling have been widely developed: G2ANet [21] employs two-stage attention with graph neural networks; DICG [22] leverages self-attention to learn deep implicit coordination graphs; DGN [43] uses multi-head attention to extract and aggregate relational features; and HAMA [44] employs hierarchical graph attention to model inter-agent and inter-group relations. Other techniques draw on relational reasoning or social psychology principles. For instance, methods use Variational AutoEncoder (VAE) or attention to infer entity-entity relationships that capture interactions among agents and environment objects [23,45], while Neighborhood Cognitive Consistency (NCC) [24] aligns nearby agents' representations via graph convolutions and variational inference to encourage cooperative behaviors.

A growing body of work applies information-theoretic principles to regularize coordination in both communication and relation modeling. Nearly Decomposable Q-functions (NDQ) [46] minimize communication through mutual information and entropy regularizers that encourage concise, task-relevant message exchanges. Information-theoretic Policy Gradient framework (InfoPG) [47] implicitly maximizes a lower bound on mutual information among the policies of cooperating agents during learning by conditioning each agent's policy on its neighbors' policies. Progressive Mutual Information Collaboration (PMIC) [48] further leverages progressive mutual information maximization and minimization to distinguish high- from low-quality collaborative behaviors. More recent efforts, such as Role-Oriented multi-agent Communication (ROCO) [49] and communication-constrained priors with dual mutual information estimation [50], explore other designs of information-theoretic regularization to address diverse communication challenges in MARL, offering principled ways to balance informativeness and efficiency.

### 9.3. Modeling behavioral diversity and agent heterogeneity

Recent MARL research has increasingly explored behavioral diversity and agent heterogeneity to promote effective role differentiation and coordinated contributions toward shared goals. Behavioral diversity is typically quantified through metrics that capture variation in agents' policies or actions. McKee et al. proposed the Expected Action Variation (EAV) to measure behavioral diversity based on sampled action distributions [25], while Hu et al. defined the Role Diversity metric from action, trajectory, and contribution perspectives to better understand the characteristics of MARL tasks [51]. Bettini et al. [26] proposed System Neural Diversity (SND) to measure the distance between stochastic policies, serving as a diversity metric. Building on SND, the Diversity Control (DiCo) method [52] was proposed to constrain the strategy learning to a target diversity level by integrating the SND into the policy architecture.

Agent heterogeneity addresses role specialization more directly. Heterogeneous Agent Trust Region Policy Optimization (HATRPO) and Heterogeneous Agent Proximal Policy Optimization (HAPPO) [27] extend TRPO and PPO to heterogeneous settings by introducing multi-agent advantage decomposition and sequential policy update with heterogeneous policy networks to enhance coordination among agents. A subsequent work generalizes this idea within the Heterogeneous Agent Mirror Learning (HAML) framework [29] and applies it to several MARL algorithms. Stateful Active Facilitator (SAF) [28] maintains a shared knowledge base and policy pool to improve knowledge exchange while encouraging strategy heterogeneity among agents. Scalable and Heterogeneous Proximal Policy Optimization (SHPPO) [53] introduces heterogeneous layers in policy networks to better accommodate differing agent capabilities and roles.

### 9.4. Key differences from previous related work

In contrast to early methods that predefine static and unweighted undirected coordination graphs [11,38], factorize the value function without explicitly defining the coordination structures [12,13], and design communication protocols [17,18], our work builds on dynamical agent relation modeling to learn implicit coordination structures and to utilize integrated representations obtained from agents' observations to better achieve collective goals in cooperative MARL.

While existing approaches involving relation modeling often infer dynamic and weighted coordination structures using proximity-based metrics, such as Euclidean distance between agent positions [20], neighborhood cognitive consistency [24], or the general attention mechanism like dot-product attention [15,22], they typically assume that agents with similar observations/perspectives cooperate more closely. Our work diverges by explicitly incorporating both consistency and discrepancy information into the learning of coordination structures. This allows our approach to capture diverse coordination dynamics, leading to more effective policy learning in complex cooperative scenarios.

A related but distinct line of methods fosters coordination by explicitly encouraging or enforcing differences among agents either at the policy/action level (e.g., via diversity objectives) or at the agent/role level (e.g., via heterogeneous policy network designs). In contrast, our method operates at the observation level and does not explicitly assume or enforce any form of behavior diversity or role heterogeneity. Instead, our DiaCoG extracts task-relevant relational signals directly from agents' observations to infer the implicit coordination relationships, which are then used to refine the subsequent value estimation and action selection. Because it introduces no new diversity objectives, auxiliary losses, or additional policy architectures, the proposed approach can be integrated seamlessly as a plug-and-play module across different base MARL algorithms and tasks.

## 10. Conclusion and future work

In this paper, we introduce DiaCoG, a novel framework that synergizes consistency and discrepancy information to construct implicit coordination graphs, so as to strengthen agent collaborations in cooperative environments. Building upon this foundation, we devise two tailored approaches, DiaCoG-DE and DiaCoG-CE, aligned with the CTDE and CTCE architectures of actor-critic networks, respectively. Both theoretical analyses and experimental assessments demonstrate DiaCoG's ability to enhance coordination dynamics, driving agents toward collective success. Additionally, the empirical validation in the SMAC environment highlights the adaptivity of the proposed DiaCoG across different MARL paradigms: it seamlessly integrates with a SOTA value-based approach and outperforms several representative and SOTA baselines. A detailed case study in the Predator-Prey environment further illustrates the mechanical improvement DiaCoG brings to coordination strategies.

We evaluate the proposed DiaCoG framework in several representative cooperative MARL environments with diverse tasks, including collaborative pursuit in Predator-Prey, autonomous vehicle coordination in Traffic Junction, and unit combat involving both homogeneous and heterogeneous agents in SMAC. These environments capture key challenges in real-world multi-agent systems, such as partial observability, dynamically changing numbers of active agents, and the need for effective collaboration among agents with varying capabilities. Our results highlight the promise of explicitly integrating consistency and discrepancy information from local observations to enhance coordination performance in complex and practical settings. Specifically, DiaCoG shows potential for broader AI applications involving multi-agent systems where the discrepancy information is inherent and matters, such as autonomous vehicle fleets with heterogeneous sensors, human-AI collaborative teams, and heterogeneous Unmanned Aerial Vehicle (UAV) swarms.

Despite these promising results, several limitations remain that point to meaningful directions for future study. First, our current framework does not explicitly address role heterogeneity, which is often critical for effective coordination in practical scenarios. Integrating DiaCoG with mechanisms that encourage role diversity at the action/policy level could yield complementary benefits in cooperative MARL. Second, our work focuses on fully cooperative scenarios. Extending the proposed framework to competitive and mixed cooperative-competitive environments, where agents face conflicting objectives or both shared and conflicting objectives, represents an important next step. Furthermore, while our information-theoretic analysis establishes the expressiveness advantage of DiaCoG over consistency-based methods in MARL, deeper theoretical investigations into the properties of the learned implicit coordination graphs and their broader implications for MARL remain valuable directions for future study.

## CRediT authorship contribution statement

**Mutong Liu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Tiantian He:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization; **Yang Liu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization; **Jiming Liu:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization; **Yew-Soon Ong:** Writing – review & editing, Methodology.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data will be made available on request.

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